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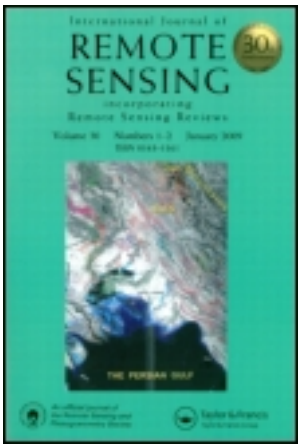

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Satellite remote sensing of submerged kelp beds on the Atlantic coast of Canada

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Satellite remote sensing of submerged kelp beds on the Atlantic coast of Canada

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Abstract. Underwater kelp seasonal variation is assessed through the comparative analysis of HRV and Thematic Mapper (TM) images of Baie des Chaleurs between Caps-Noirs and Pointe-Bonaventure, Quebec. The total biomass is estimated, based on the morphology of the dominant species *Laminaria longicruris*. Kelp-covered and kelp-free areas are differentiated from each other in water depth of 0–6 m and 0–7 m with the HRV and TM images, respectively. The median biomass estimated for the kelp-covered category of the classified image is $(1500 \pm 400)\,\mathrm{g\,m^{-2}}$. The multidate image shows a spatial variation of the kelp beds in 45% of the area. Areas where no change occurred occupy at least 70 ha, while growth and decay of kelp are observed in much smaller areas, in shallow water and at the boundary of kelp beds.

Résumé. La variation saisonnières d'un peuplement de macrophytes submergés est analysée à l'aide de la comparaison d'images HRV et TM. Le site d'étude est localisé dans la baie des Chaleurs, entre Caps-Noirs et Pointe-Bonaventure, Québec. La biomasse totale est estimée à partir des caractéristiques morphologiques de l'espèce dominante *Laminaria longicruris*. Les peuplements de macrophytes submergés et le substrat nu sont différenciés dans la zone de 0 à 6 m et de 0 à 7 m de profondeur, respectivement avec les images HRV et TM. Une valeur médiane de biomasse de $(1500 \pm 400)\,\mathrm{g\,m^{-2}}$ est estimée pour la classe correspondant aux peupements d'algues submergées. L'image multidate met en évidence une variation spatiale des peuplements sur 45% de la zone d'étude. Les secteurs où aucun changement n'a été observé occupent au moins 70 ha, ce qui contraste avec les changements qui sont généralement observés sur de petites superficies en eau peu profonde et en bordure des peuplements.

1. Introduction

Data on marine kelp ecology have traditionally been obtained through field surveys, scuba diving and laboratory analyses. These methods have been combined with remote sensing to locate kelp beds, differentiate species, monitor changes and estimate the biomass. Several studies, which are summarized in Belsher *et al.* (1996) and Lambert *et al.* (1987), have been published on the topic since Cameron's (1950)

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work based on aerial photography. It is apparent from these studies that change detection and biomass estimation strategies have not yet been fully explored for submerged kelp beds. The monitoring of marine and aquatic vegetation requires that multisensor data integration be considered, for both long-term and short-term change detection. On the one hand, long-term monitoring is impeded by the short lifespan of remote sensing systems. Jensen *et al.* (1987) addressed this problem by comparing a Landsat Multi-Spectral Scanner (MSS) to a Landsat Thematic Mapper (TM) image, respectively recorded in 1974 and 1984, to detect the changes in *Macrocystis pyrifera* beds distribution. On the other hand, high temporal resolution image sets required for short-term monitoring (Augenstein *et al.* 1991, Jensen *et al.* 1993) are often inaccessible due to weather conditions. For example, in Canada's southern latitudes, cloud-free image temporal resolution is approximately 30–60 days for SPOT, and 1–2 years for TM (Clavet 1991). The weather also affects change detection that, ideally, use images recorded on an anniversary date (Zainal *et al.* 1993, Dobson *et al.* 1995). Finally, a single satellite optical sensor recording cycle does not address the need of emergency monitoring (Klemas 1984, Jensen *et al.* 1987). Multisensor image combination increases the effective temporal resolution available for resource monitoring. Analogue spatial resolution, spectral bands and radiometric characteristics (table 1) suggest that comparable information can be extracted from HRV and TM visible images.

Intertidal and submerged kelp biomass assessments are based on airborne (Sharp *et al.* 1981, Grenier 1987, Lavoie *et al.* 1990, 1991, Sommaruga *et al.* 1993) and satellite (Armstrong 1993) data. However, accurate quantitative information is difficult to obtain because the spatial dimension of *in situ* reference data is not compatible with the information conveyed by the images. Traditionally, *in situ* data have been collected in 1 m^2 quadrats distributed at a short distance from each other, and the sampled biomass is assumed representative of an area defined by the image pixel resolution for the corresponding location. This assumption is not valid if the sampling quadrat size is different from the image spatial resolution (Congalton and Green 1999). The collection of *in situ* biomass data over a sampling area equal to the pixel resolution as proposed by Belsher *et al.* (1996) is not a realistic solution with the satellite images currently available. The principal objective of this study is to assess underwater kelp seasonal variation through the comparative analysis of HRV and TM images.

The secondary objectives are as follows: (1) to assess the water depth at which kelp beds and bare substrate can be differentiated from each other on the HRV and TM images, and (2) to produce a non-site specific kelp biomass estimate. The non-site specific approach—namely, not attempting to match a sampling quadrat to any

Table 1. Characteristics of HRV and TM visible bands.

	HRV		TM		
Channel	1	2	1	2	3
Spectral band	green	red	blue	green	red
Centre wavelength (nm)	545	645	485	560	660
Band width (nm)	90	70	70	80	60
Radiometric resolution (NE$\Delta\rho$)	0.005	0.005	0.008	0.005	0.005
Spatial resolution (m)	20			30	

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specific image pixel—is proposed because it minimizes the problem of different spatial resolutions between the field data and the satellite images.

2. Study area and data

The study area includes 30 km of coastline on the north coast of the Baie des Chaleurs, between Caps-Noirs and Pointe-Bonaventure, in Quebec (figure 1). Previous field inventories (Cardinal 1967, Gendron 1984) and aerial photograph interpretation (Lavoie *et al.* 1985) reported important variations in the distribution and biomass of kelp beds. The species composition is dominated by *Laminaria longicruris* (Gendron 1984, Gendron, unpublished), and the majority of undercanopy species includes *Fucus* sp. and *Chondrus crispus*. The kelp resources of the Baie des Chaleurs have been used to fertilize local crops, but intensive harvesting has not taken place in the past.

Data used in this study include scuba diving inventories, *in situ* reflectance data, and satellite images recorded by the Landsat TM and SPOT HRV multispectral sensors. Scuba diving inventories conducted in 1980 and 1981 included information on water depth and biomass of *Laminaria longicruris*. The water depth was reported according to 2 m class intervals and the kelp biomass expressed in g m^{-2} wet weight unit. Over 300 quadrats of 1 m^2 along 21 transects lying perpendicular to the coastline were examined by Gendron (1984). The transects were 1–3 km away from each other, varied from 150 to 2.3 km in length, and the quadrats were 50–100 m apart.

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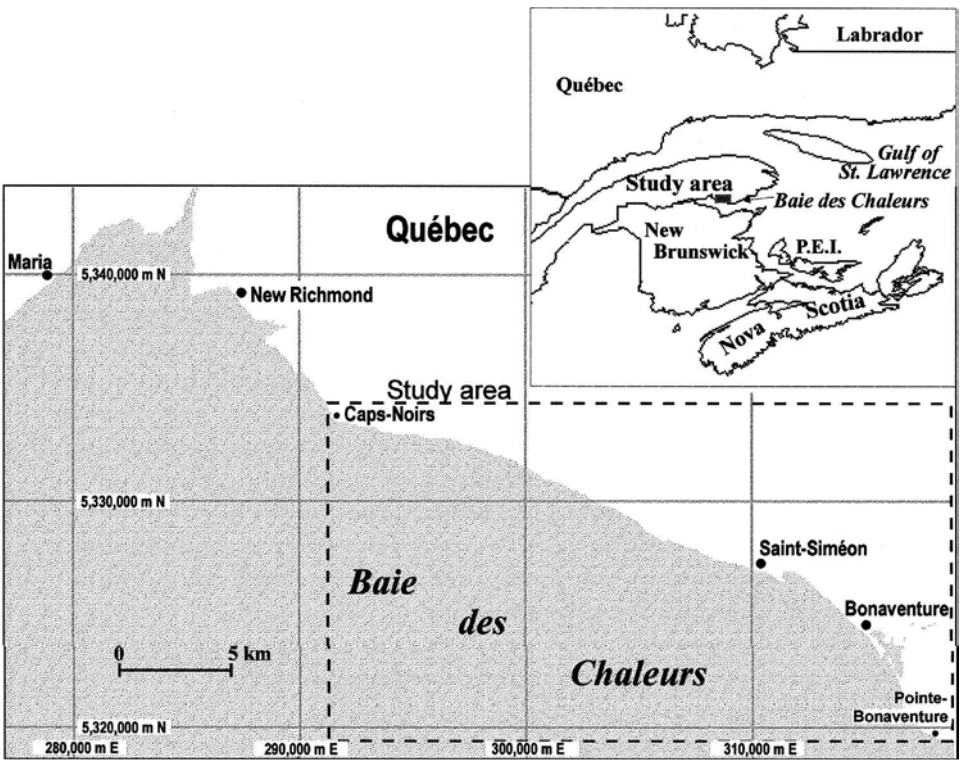

Figure 1. Study area location map. The study area includes 30 km of coastline on the north coast of Baie des Chaleurs, between Caps-Noirs and Pointe-Bonaventure.

In situ reflectance data were recorded with a Spectron SE590 portable spectroradiometer working in the 378–1114 nm wavelength range. Spectral response curves obtained for sand and kelp bottom types were analysed by Gauthier (1991) to simulate the reflectance of kelp submerged under variable water depths.

The HRV and TM images used in this study were recorded on 10 October 1986 and 16 July 1987, respectively. Both images were geometrically corrected to the UTM (zone 20) reference system using their particular spatial resolution (20 m for HRV and 30 m for TM). The HRV image was then resampled to the TM image pixel size 30 m before producing the coincidence matrix for multidate analysis. The 30 m resolution was chosen bearing in mind that resampling an image to a finer resolution introduces spatial autocorrelation which, in turn, may bias the outcome of coincidence matrix analysis (Anselin and Getis 1993, Congalton and Green 1999).

Atmospheric corrections and radiometric calibrations were applied using the '5S-inverse' model (Tanré *et al.* 1986, Royer *et al.* 1988). As a result, the image digital numbers (DNs) represented the surface's apparent reflectance.

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3. Methods

In situ data provided the base to assess reflectance difference between kelp-covered substrate (100% kelp cover) and bare substrate (0% kelp cover) at various water depths. The limit radiometric resolution of the sensor is set by the noise equivalent reflectance (NE$\Delta\rho$). The 'detectability limit' corresponds to the water depth at which the difference between the reflectance of kelp-covered and bare surfaces is equal or smaller than the NE$\Delta\rho$. Analyses considered the attenuation coefficient of Type 1 coastal water (Jerlov 1976). Secchi disk measures (S) collected in the study area were used in the equation

$$k_{400\text{–}500\text{nm}} = (1.6 \pm 0.6)/S \tag{1}$$

to estimate the blue band attenuation coefficient ($k_{400\text{–}500\text{nm}}$) (Pickard and Emery 1982). The result, $k_{400\text{–}500\text{nm}} = 0.228 + 0.008$, corresponds to the Type 1 water. Consequently, $k_{550\text{nm}} = 0.1415$ and $k_{650\text{mn}} = 0.4525$ (Austin and Petzold 1984) were applied for estimating the submerged kelp and bare substrate reflectance values.

The logarithmic transformation of a set of visible bands resulted in N water depth dependent variables. The linear combination of X_j images produces the water depth image (Y_N), and the orthogonal produces the bottom type image (Lyzenga 1978). The bottom type image (Y_i) is calculated as follows:

$$Y_i = \sum_{j=1}^{N} A_{ij} X_j \tag{2}$$

where A = transformation parameter, X = variable linearly correlated with water depth, and j = spectral band.

Several methods have been proposed to rotate the coordinate system of the X_j images set with the objective to approximate the effect of water attenuation on the bottom multispectral reflectance. The application of the method is simplified by the principal component analysis (PCA) transformation as suggested by Khan *et al.* (1992). However, with this method it is assumed that the kelp canopy does not modify the water depth. In addition, the rotation coefficients include radiance values that are not water depth dependent; rather they represent the intertidal zone and very deep water. The inclusion of these factors, in a PCA, reduces the potential of the transformed image to extract information representative of the bottom type.

The method proposed in this study is based on the discriminant analysis of the dependent X_j image set. The coefficients applied to produce the images Y_N and Y_i are derived from a small image subset in which X_j values are strongly correlated with the water depth variation. A high correlation insures that the first discriminant axis (Y_N) represents the water depth and, therefore, the second discriminant function (Y_i) depicts the bottom type.

A digital elevation model (DEM) provided the water depth information. The homogeneity of the bottom type was verified through visual interpretation of the image and field data. It was important to exclude kelp-covered substrate from the subset because the plant's vertical structure hides the actual water depth variation. The subset excluded reflectance values that are not depth dependent. These would represent, for example, deep water (as defined by the detectability limit of the sensor in use), very turbid water, surface features such as vessels and wharves, or clouds. The coefficients describing Y_N must essentially represent water depth variations in order to obtain a second discriminant axis Y_i that contains bottom type related variation.

A calibrated equation represents the output image with the desired range of values; in this study an 8-bit image (256 DNs) was used. The resulting image, classified through density slicing, depicts information that primarily relates to the kelp beds' canopy coverage. A group of pixels that have a normal distribution defines a particular bottom type (Lyzenga 1978). Each class formed through density slicing is evaluated for the probability ($p(n,n)$) that the class average digital number represents its respective distribution. The density function probability is calculated from the equation

$$p(n,n)=f^*[\,Y_{n,\min}-Y_{n,\mu}\}/\sigma_n\}+f^*\{Y_{n,\max}-Y_{n,\mu}\}/\sigma_n\} \tag{3}$$

where f^* = distribution function coefficient (Scherrer 1984: Appendix A, table III), Y_i = bottom type image, n = bottom type category, min = minimum DN, max = maximum DN, μ = average DN and σ = standard deviation.

Multidate variation of the kelp canopy coverage is given by the overlay of two thematic images, after being resampled to a common spatial resolution. A coincidence matrix shows the changes of canopy coverage that occurred in the period between October 1986 and July 1987. The multidate image displays four categories as follows: (1) no change in kelp-free area, (2) decrease in kelp coverage, (3) increase in kelp coverage, and (4) no change in kelp-covered area.

The biomass was derived for the categories of the classified images by matching them with the 'total wet weight' data collected in the study area (note: the field data were collected for biological studies that, at the time, were not intended to be used with remote sensing images). The analysis uses *in situ* data collected without geographic coordinates, but they were acquired in the same area as the image. The relative area coverage represented in the thematic image categories was linked to the absolute area coverage within one sampling unit. Given that the absolute area coverage is due to at least one plant, its morphology is used to approximate the absolute leaf area by multiplying the blade length and width. *Laminaria longicruris* dimensions are derived using the blade width to length ratio of 1:5 (Gendron 1984, Williams 1987). The leaf area of one plant can therefore be expressed as a percentage of 1 m^2 sampling quadrats.

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The plant's blade length (BL in cm) is related to the total weight (TW in g) using the relationship

$$BL^{1.809} = 43.48\,BW = 23.91\,TW \tag{4}$$

which accounts for the fact that the blade weight (BW in g) represents on average 55% of the total weight of a species (Gendron 1984).

The biomass values defining the class limits indicated a threshold to isolate two groups of biomass values. Then, each group was characterized in a 'letter-value display' that reports on a median value, extreme values and range, which are representative within 95% confidence level (Velleman and Hoaglin 1981).

4. Results and discussion

The *in situ* reflectance data confirmed the similar capabilities of TM and HRV images to detect underwater kelp beds. The differentiation of 100% kelp and kelp-free substrates can be applied to water depths of 0–6 m with the HRV image and 0–7 m with the TM image (figure 2).

Table 2 reports the correlation coefficients between the water depth and the water depth dependent images (X_j) for seven locations. The image subset no. 6 produced the highest correlation coefficient between the water depth and all X_j variables. At that location was also found one of the highest average reflectance values, indicating that the substrate is mainly macrophyte-free. The attenuation of blue (X_{TM1}) and red (X_{HRV2} and X_{TM3}) light in coastal water (Jerlov 1976) is revealed by the low correlation coefficient in these bands. As expected, the green band (X_{HRV1} and X_{TM2}) provides the highest potential to discriminate between underwater bottom types.

The coefficients of the second discriminant function are derived from the values included in the subset no. 6 that have the highest correlation coefficient with water

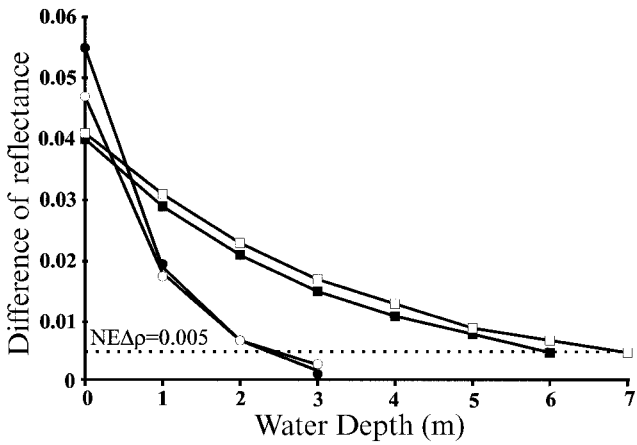

Figure 2. Difference of reflectance between kelp-free and kelp-covered surfaces in HRV and TM image bands. The two bottom types can be differentiated from each other if the difference of reflectance is greater than the noise equivalent reflectance (NE$\Delta\rho$). For example, the kelp-free and kelp-covered surfaces can be differentiated at water depths of 0–7 m with the TM band 2. ■, HRV band 1 (green) simulation; ●, HRV band 2 (red) simulation; □, TM band 2 (green) simulation; ○, TM band 3 (red) simulation.

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Table 2. Correlation between water depth and water depth dependent image for seven samples selected across the study area. The highest correlation values were obtained with sample no. 6.

Image:		HRV			TM		
		Spectral band			Spectral band		
Sample no.	Area (ha)	Green	Red	Area (ha)	Blue	Green	Red
1	22	−0.35	−0.58	27	−0.05	−0.24	−0.49
2	7	−0.27	−0.31	22	+0.40	+0.32	+0.05
3	7	−0.10	−0.18	17	+0.03	+0.02	−0.10
4	8	−0.78	−0.83	36	+0.05	−0.15	−0.42
5	45	−0.82	−0.76	50	−0.33	−0.58	−0.64
6	30	−0.88	−0.86	41	−0.59	−0.74	−0.73
7	17	−0.54	−0.55	22	+0.29	+0.15	+0.01

depth variation. The transformation equations applied with the HRV (equation (5)) and TM (equation (6)) images are as follows:

$$BT_{XS} = 0.87X_1 - 0.50X_2 \tag{5}$$

$$BT_{TM} = 0.36X_1 + 0.83X_2 - 0.43X_3 \tag{6}$$

where BT_{HRV} = digital number of the bottom type HRV image, BT_{TM} = digital number of the bottom type TM image, X = digital number of the water depth dependent image, and j = spectral band.

The bottom type image values range from 0 to 255. Two categories have been identified through density slicing classification; they correspond to 0–50% kelp cover and 51–100% kelp. A normal distribution function represents each category ($\alpha = 0.01$). In addition, the probability of good classification is high for most classes, ranging from 0.77 to 0.99 (table 3).

Table 4 presents the biomass letter-value display for each category of the thematic images. The biomass significant figure is $(10 \pm 10)\,\mathrm{g\,m^{-2}}$ for class C (0–50%). A relatively low density characterizes the kelp population in class C (51–100%) with a biomass of $(1500 \pm 400)\,\mathrm{g\,m^{-2}}$. The latter value is nearly identical to the average biomass estimated by Gendron (1984) in the area delimited by Rivière Caplan and Caplan. However, the biomass values obtained for *L. longicruris* 100% canopy coverage, at other locations, are generally higher than the value derived from the plant morphology. Surveys on the lower Quebec north shore showed biomass values

Table 3. Probability that the class average represents the category defined through density slicing. The probability values were calculated for the bottom type image distribution functions corresponding to kelp-free (C(0–50%)) and kelp-covered (C(51–100%)) categories.

HRV image		TM image	
C(0–50%)	C(51–100%)	C(0–50%)	C(51–100%)
0.89	0.99	0.77	0.89

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Table 4. Letter-value display of biomass values. One median value and three sets of boundary values (hinge, outer fence and 95% confidence interval) represent each bottom type category (C). The spread is defined by two boundary values. A boundary value is undefined if less than 0.

	Lower	Upper	Spread
C(0–50%), n = 96; biomass estimated from plant morphology: 0–397 g m^{-2}			
Median		6	
Hinge	0	64	64
Outer fence	0	191	191
95% confidence interval	0	16	16
C(51–100%); n = 54; biomass estimated from plant morphology: > 406 g m^{-2}			
Median		1484	
Hinge	746	1854	1108
Outer fence	–	6785	–
95% confidence interval	1104	1864	760

of 2.3–5.7 kg m^{-2} (DFO 1991). Values ranging from 0.9 and 3.6 kg m^{-2} were reported on the south shore of Newfoundland for *L. longicruris* (Simms 1997).

The biomass in the area of Caps-Noirs to Pointe-Bonaventure in total is, for the 0 to 6 m water depth zone, estimated to (13 000 ± 3000) metric tons (t) on both of the images. This value exceeds the scuba diving estimate of (9000 ± 2000) t obtained for the corresponding water depth categories (Gendron 1984). The difference between the total biomass estimates is explained by the generalization introduced in the scuba diving area calculations. The 0–6 m water depth area covers 1920 ha on the DEM and covers approximately 1300 ha in the scuba diving assessment.

The coincidence matrix comparing the HRV and the TM thematic images was calculated for the 0–6 m water depth zone. Table 5 presents the analysis result, which exhibits a 19 ha decrease in the kelp population between October 1986 and July 1987. The corresponding biomass loss is (280 ± 70) t. However, this value is only indicative of a biomass loss because it is inferior to the interval of variation of estimates on individual images. Areas of 876 and 857 ha belong to C(51–100%) of the HRV and TM images, respectively. The spatial distribution of kelp beds is similar on both images. Kelp beds depicted on the HRV image (figure 3(*a*)) are close to the coastline between Caps-Noirs and Saint-Siméon, and in the vicinity of Pointe-Bonaventure. On the TM image (figure 3(*b*)), they are 50–100 m offshore along the entire coastline.

Pixel-specific comparison (table 5) revealed that spatial variation (increase or decrease in biomass) occurred in 45% of the area. The multidate image (figure 3(*c*))

Table 5. Coincidence matrix of the kelp area coverage (ha) changes resulting from the comparison of the October 1986 HRV image and the July 1987 TM image.

July 1987 TM image	October 1986 HRV image		Total
	C(0–50%)	C(51–100%)	
C(0–50%)	620	443	1063
C(51–100%)	424	433	857
Total	1044	876	1920

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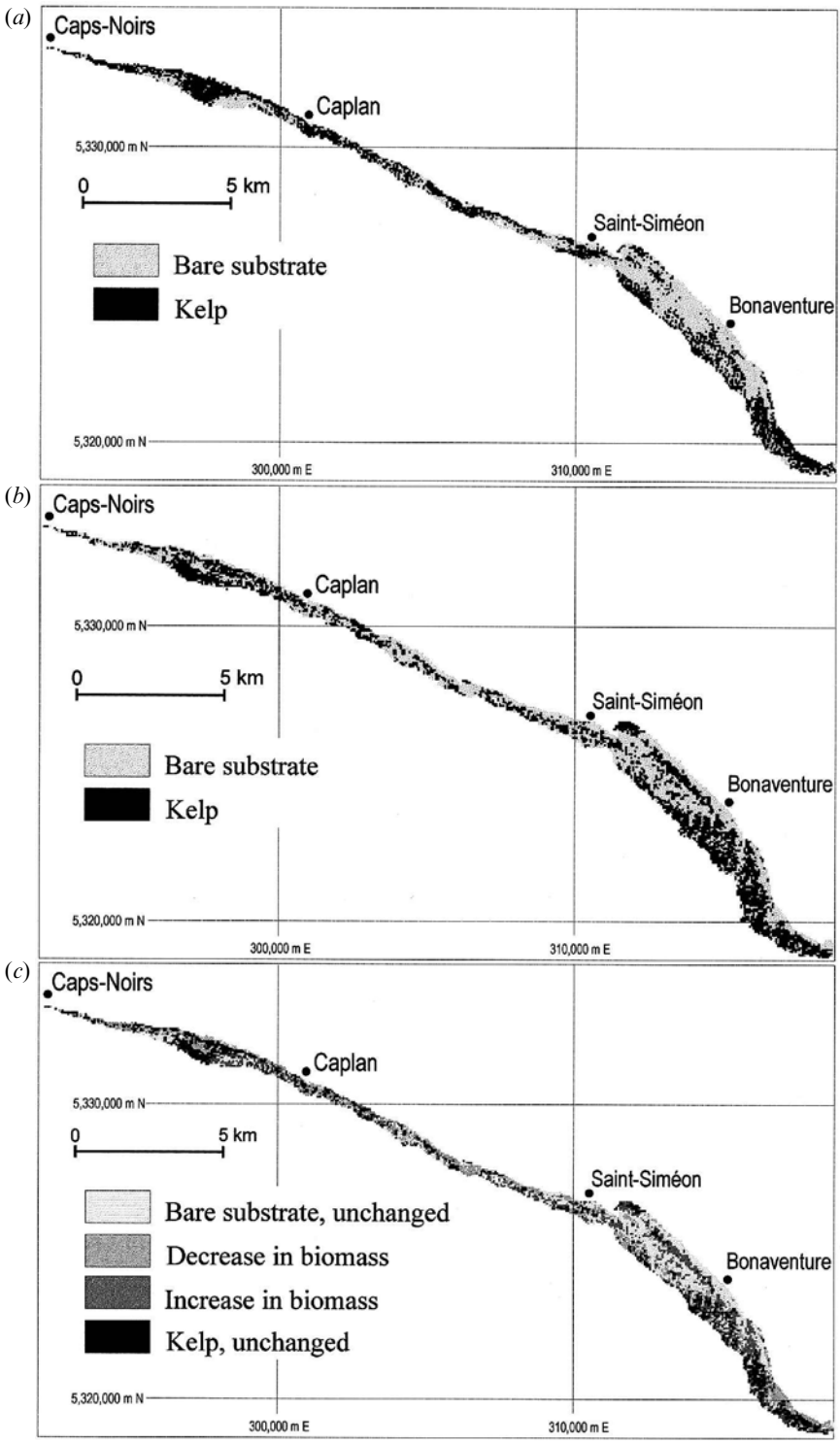

Figure 3. Bottom type HRV (*a*) and TM (*b*) images depict the bare substrate and kelp-covered substrate classes. The multidate image (*c*) illustrates the spatial distribution of changes that occurred between October 1986 (HRV image) and July 1987 (TM image).

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shows that the loss of biomass generally occurred close to the coastline, and the kelp population has expanded at a distance from the shore. The opposite is noticed in the area of Saint-Siméon to Bonaventure. In that area the underwater terrain slope is much less abrupt than in the rest of the study area, and shallow water kelp beds' growth is concentrated in a narrow band detached from deeper water kelp beds. The categories where no changes have occurred occupy by far the largest contiguous areas. The first class, indicating no change of C (0–50%) extends over 327 ha between Saint-Siméon and Bonaventure, and the second class, representing no change of C (51–100%) occupies a continuous surface of 79 ha, 10 km east of Caps-Noirs. The largest areas characterized by increased kelp coverage are in the vicinity of Bonaventure, where two groups form a total area of 91 ha. The largest contiguous areas of decreased kelp coverage are found near the coastline West of Pointe-Bonaventure and West of Saint-Siméon; each of them covers approximately 40 ha.

The multidate image suggests that the kelp beds remained stable during the Fall and Winter seasons, in the 0–6 m water depths. The growth and decay of the populations occurred in small areas of shallow water or on the edges of stable kelp beds.

5. Conclusion

Two types of satellite images were used to assess the seasonal changes in the distribution and biomass of submerged kelp beds. The information was obtained for water depth of 0–6 m with the HRV multispectral image and 0–7 m with the TM image. The multitemporal analysis showed a net 19 ha decrease in the area covered by the kelp beds. A total biomass of 13 000 t was estimated for the 0–6 m water depth over a distance of 30 km. The biomass estimate was based on the morphology of the plants. This method may result in an underestimation of the biomass. However, the non-site specific approach used allowed to make a correspondence to be established between the biomass and the image without having to sample *in situ* data over areas as large as the image spatial resolution.

An appropriate *in situ* dataset would, in addition to the biomass (wet or dry weight), include the canopy coverage (in %) and the geographic location of quadrats. The relationship between the *in situ* biomass and canopy coverage values would readily provide a biomass estimate for the kelp-cover classes obtained with the satellite image. HRV and TM images provide a similar quality of information. The combination of these two types of images is a viable solution for natural resources monitoring. It addresses the problems caused by the difference between the temporal resolution requirements of change detection, and the recording cycle of a single satellite sensor.

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